Does Female Leadership Relate to Productivity-Wage Gaps in Colombia?

Applied Microeconomic Theory ECON 6301 Paper

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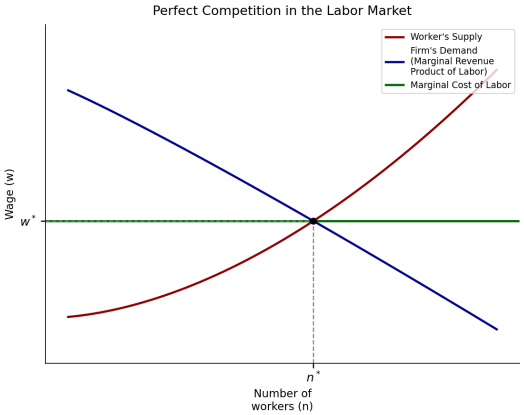
Introduction

- Across the world, workers are paid substantially less than the value they produce. For example, estimates for Colombia suggest workers produce approximately 40% more than they earn.
- Microeconomic theory explains this productivity-wage gap (markdown) through monopsony power in the labor market.
- Labor markdowns represent not only foregone income for workers, but also a significant source of allocative inefficiency for the broader economy.
- Although monopsony power arises from structural features of labor markets, firms differ substantially in how aggressively they exploit it.
- Literature suggests that female leaders place greater weight on worker welfare and are less likely to engage in anticompetitive behavior.

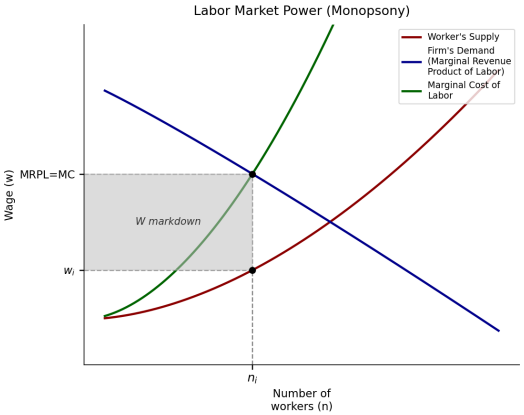
This paper examines whether greater female leadership reduces the extent to which firms exploit monopsony power, i.e., whether it is associated with a lower productivity-wage gap.

Theoretical Framework

Perfect Competition



Monopsony



Data

- Firm-level data from the Annual Manufacturing Survey (Encuesta Anual Manufacturera, EAM) of Colombia.
- 11,235 Colombian manufacturing firms observed over 2000–2019, yielding 53,688 firm-year observations spanning 105 industries and 23 geographic regions.

Table: Descriptive Statistics

Variable	Mean	SD	Min	Median	Max
Share of female executives (%)	45.192	39.925	0.000	49.812	100.000
Firm's share of industry employment (%)	0.020	0.066	0.000	0.004	1.000
Share of workers in production (%)	61.951	21.743	0.000	66.667	100.000
Gross output ($ millions USD)	1.219	536.010	756.959	11.356	2,968.214
Firm age (years)	4.262	3.422	1	3	20
Number of workers	56	146	1	20	4921
Wage per worker ($ thousands USD)	1.827	91	349.180	2.296	1,498.014

Markdown definition

Firm's profit maximization under monopsony:

$$\max_{L} \Pi = R(L) - \bar{w}L, \tag{1}$$

Inverse Elasticity Pricing Rule for a Monopsonist:

$$\frac{R'(L) - w(L)}{w(L)} = \frac{w'(L)L}{w(L)} \equiv \varepsilon_L^{-1} \tag{2}$$

Rearranged following Yeh et al. (2022) we obtain the firm's markdown:

$$\nu \equiv \frac{R'(L)}{w(L)} = \varepsilon_L^{-1} + 1 \tag{3}$$

Markdown estimation - Production Approach

Following Yeh et al. (2022) and using the production approach by De Locker et al. (2020) we can obtain the firm's markdown as:

$$\nu = \frac{\theta^L}{\theta^M} \cdot \frac{p^m M}{w(L)L}. \tag{4}$$

Where θ^L and θ^M are extracted from a translog production function GMM estimation:

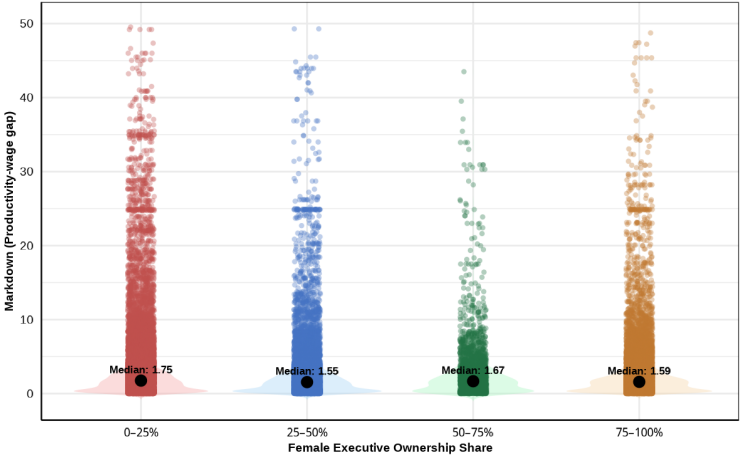
$$\begin{aligned}\log(GO_{it}) = {} & \beta_0 + \beta_1 \log(K_{it}) + \beta_2 \log(L_{it}) + \beta_3 \log(M_{it}) + \beta_4 (\log K_{it})^2 + \beta_5 (\log L_{it})^2 \\ & + \beta_6 (\log M_{it})^2 + \beta_7 \log(K_{it}) \log(L_{it}) + \beta_8 \log(K_{it}) \log(M_{it}) + \beta_9 \log(L_{it}) \log(M_{it}) \\ & + \gamma_t + \varepsilon_{it}.\end{aligned} \tag{5}$$

As follows:

$$\begin{aligned}\hat{\theta}_{L,it} &= \hat{\beta}_2 + 2\hat{\beta}_5 \log(L_{it}) + \hat{\beta}_7 \log(K_{it}) + \hat{\beta}_9 \log(M_{it}), \\ \hat{\theta}_{M,it} &= \hat{\beta}_3 + 2\hat{\beta}_6 \log(M_{it}) + \hat{\beta}_8 \log(K_{it}) + \hat{\beta}_9 \log(L_{it}).\end{aligned} \tag{6}$$

Markdown and Female Leadership

Figure: Markdown and Female Leadership



Empirical Framework

Linear regression model:

$$\log(\text{Markdown}_{it}) = \alpha + \beta\,\text{FemaleLeadership}_{it} + \boldsymbol{\gamma}'\mathbf{X}_{it} + \alpha_t + \alpha_s + \alpha_r + \varepsilon_{it} \tag{7}$$

where:

- $\log(\text{Markdown}_{it})$ is the logarithm of the productivity-wage gap for firm i in year t
- $\text{FemaleLeadership}_{it}$ denotes the share of women in executive and ownership positions
- The vector $\mathbf{X}_{it}$ includes a rich set of firm-level controls designed to isolate the leadership channel from confounding variation in productivity, scale, and workforce composition.
- α_t, α_s, and α_r are year, industry, and region fixed effects.
- ε_{it} is the error term.

Regression Results

Table: Regression Results - Dependent variable: log(Markdown)

	(1)	(2)	(3)
Share of Female Executives	-0.0010***	-0.0010***	-0.0011***
	(0.0001)	(0.0002)	(0.0002)
Firm's share of employment in the industry		2.0356***	2.4268***
		(0.3723)	(0.3713)
Share of production workers		-0.0021***	-0.0018***
		(0.0003)	(0.0003)
Industry HHI		0.5729***	0.5719***
		(0.0634)	(0.0635)
TFP		0.1557***	0.1578***
		(0.0039)	(0.0039)
Additional controls		Size-Age	Size-Age
FE: Year	X	X	X
FE: Region	X		X
FE: Industry	X		
N	48,292	31,664	31,664
R^2	0.5097	0.1243	0.1386

Note: $^{***}p < 0.001$; $^{**}p < 0.01$; $^{*}p < 0.05$; $^{\cdot}p < 0.1$

Conclusion

- Management structure is a meaningful margin of variation in firms' wage-setting behavior.
- A larger share of female executives is associated with less aggressive exploitation of monopsony rents.
- This is consistent with evidence that female leaders place greater weight on worker welfare and interpret productivity signals differently from their male counterparts.
- Policies promoting gender diversity in executive and ownership positions (board gender quotas, bias-free promotion practices, and mentorship pipelines) may generate positive labor market externalities.